2. A Heterogeneous Sensing System for Soil Moisture Mapping in Agricultural Environments (2025)

Year: 2025

Publisher: Computers and Electronics in Agriculture

Research Area: Advanced Smart Farming Technologies for Agricultural Automation

Why it is Relevant:

Relates to agricultural automation and supports smart systems for monitoring and irrigation management. The study combines multiple sensing technologies and autonomous platforms to improve soil moisture monitoring and precision agriculture applications.

Summary:

The paper presents a heterogeneous sensing framework that integrates wireless sensor networks, autonomous mobile robots, and portable soil moisture probes to generate high-resolution soil moisture maps. Data collected from different sensing sources are fused using advanced machine learning and Gaussian Process Regression techniques. The system provides accurate field-scale moisture distribution with uncertainty estimation, helping farmers make better irrigation decisions. Experimental results demonstrate that combining multiple sensing platforms improves coverage, mapping accuracy, and overall efficiency compared to using a single sensing method.

Relation to Your Project:

• Multi-sensor integration ✔

• Autonomous robotic monitoring ✔

• Soil moisture mapping ✔

• Precision irrigation support ✔

• Agricultural automation ✔

• Smart farming decision-making ✔

• Data fusion and machine learning analysis ✔